

1 Linear Combinations

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^2 (a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^1 (a_u^r)^1}_{a_g^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^1 (a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^2 (a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{a_g^r} \right|$$

$$+ \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{a_g^r} \right|$$

dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\
& +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\
& +2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\
& +2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$+ \quad 1 \quad \cdot \quad |(b_{1u}^r) (b_{2g}^r) (b_{3g}^r) (a_u^r)| \quad + \quad 1 \quad \cdot \quad |(b_{1u}^l) (b_{2g}^l) (b_{3g}^l) (a_u^l)|$$

substitution by dimer:

$$+1 \cdot |(-a_g + b_{1u}) (-b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{2u} + b_{3g}) (+b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| \\
& \quad + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
& \quad + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right|
\end{aligned}$$

$$S * Q - Q * S$$

$$+ \begin{pmatrix} \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \downarrow\uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\uparrow\uparrow} & \overline{\uparrow\uparrow} \\ \uparrow & \uparrow & \downarrow\uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^2 (a_u^l)^0}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^1 (a_u^r)^1}_{a_g^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^1 (a_u^l)^1}_{a_g^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^2 (a_u^r)^0}_{a_g^r} \right|$$

substitution by dimer:

$$+ \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{a_g^r} \right|$$

$$- \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{a_g^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{a_g^r} \right|$$

dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g}
\end{array}$$

summed up:

$$\begin{aligned}
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} - 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1u}} - 2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{b_{1u}} \\
& -2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \\
& -2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}}
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$+ 1 \cdot |(b_{1u}^T) (b_{2g}^T) (b_{3g}^T) (a_u^T)| - 1 \cdot |(b_{1u}^L) (b_{2g}^L) (b_{3g}^L) (a_u^L)|$$

substitution by dimer:

$$+1 \cdot |(-a_g + b_{1u}) (-b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{2u} + b_{3g}) (+b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \\
& + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \\
& - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \\
& \quad - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}
\end{aligned}$$

summed up:

$$\begin{aligned}
& -2 \cdot \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - 2 \cdot \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - 2 \cdot \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - 2 \cdot \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - 2 \cdot \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - 2 \cdot \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \\
& \quad - 2 \cdot \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - 2 \cdot \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}
\end{aligned}$$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right|$$

substitution by dimer:

$$\begin{aligned} & + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ & - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\ & - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\ & + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \end{aligned}$$

dimer ci vectors:

$$\begin{aligned}
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g}
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$+1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{3g}^r)| - 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| - 1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| + 1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| \\
& + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|
\end{aligned}$$

$$i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

summed up:

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|$$

or, expressed in terms of symmetry:

monomer determinants:

$$+1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{2g}^r) (a_u^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\ -1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| - 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| + 1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\ +1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| + 1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\ -1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| - 1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\ -1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| - 1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} + \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \\
& - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} + \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} + \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \\
& + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \\
& - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} + \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \\
& - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \\
& + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} + \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \\
& - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \\
& + \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \\
& - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} + \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& - \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} - \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}
\end{aligned}$$

summed up:

$$+ \quad 2 \quad \cdot \quad \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \quad - \quad 2 \quad \cdot \quad \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}$$

$$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

summed up:

$$\begin{aligned}
& -4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| \\
& -4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| \\
& +4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| \\
& +4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{3g}^r)| - 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{2g}^r) (a_u^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| - 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

[illegible]

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \\
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} - 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} + 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} + 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u}
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| - 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& +1 \cdot |(b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \\
& + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \\
& - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

$$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& -2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} + 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} + 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \\
& +2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}}
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| - 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \\
& - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \\
& + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right|
\end{aligned}$$

$$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& - \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& - \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& - \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& - \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

dimer ci vectors:

summed up:

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\
& -2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\
& +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| \\
& + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| + \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| - \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right| + \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right| - \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right| \\
& \quad + \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right|
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

$$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} + 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} + 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}}
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& -1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \\
& - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \\
& - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right|
\end{aligned}$$

$$e^3 b_{2u} * e^3 b_{2u}$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{array} \right)$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right|$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

$$\begin{aligned}
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g}
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$+1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{2g}^r) (a_u^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| + 1 \cdot |(+a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| + 1 \cdot |(+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| \\
& - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow & \uparrow & \uparrow & \downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow & \downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow & \uparrow & \downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow & \uparrow & \downarrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow & \uparrow & \downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow & \uparrow & \downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow & \downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow & \uparrow & \uparrow & \downarrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\
& +2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \right| \\
& +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \right| \\
& +2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| - 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| \\
& + \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| - \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| + \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| + \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| \\
& - \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right| + \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right| - \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right|
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

$$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \\
& + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& -2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} + 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& +2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} + 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}}
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| - 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| - 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \\
& - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \\
& - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \\
& - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right|
\end{aligned}$$

$$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^0}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^0}_{b_{2u}^r} \\
& + \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^1(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^0}_{b_{3u}^r} \\
& + \underbrace{(+a_g + b_{1u})^1(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^0}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^1}_{b_{2u}^r} \\
& + \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^1(+b_{2u} + b_{3g})^2(+b_{1g} + a_u)^1}_{b_{2u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^2(-b_{2u} + b_{3g})^1(-b_{1g} + a_u)^1}_{b_{3u}^r} \\
& + \underbrace{(+a_g + b_{1u})^0(+b_{3u} + b_{2g})^2(+b_{2u} + b_{3g})^1(+b_{1g} + a_u)^1}_{b_{3u}^l} \cdot \underbrace{(-a_g + b_{1u})^0(-b_{3u} + b_{2g})^1(-b_{2u} + b_{3g})^2(-b_{1g} + a_u)^1}_{b_{2u}^r}
\end{aligned}$$

dimer ci vectors:

summed up:

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_u} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_u} \\
& -2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_u} + 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_u} \\
& +2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_u} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_u} \\
& -2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_u} + 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_u}
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& +1 \cdot |(b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \\
& + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \\
& + \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} - \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \\
& - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} - \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + 2 \cdot \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$

$$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^0}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^1}_{b_{2u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^0}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^0}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^1}_{b_{2u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^1}_{b_{2u}^r} \right|
\end{aligned}$$

dimer ci vectors:

$$\begin{array}{l}
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_u} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
- \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_u} \\
- \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1g}} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} \\
- \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1g}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_u} \\
+ \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_u} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1g}}
\end{array}$$

summed up:

$$\begin{aligned}
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& +2 \cdot \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1g}} \\
& +2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}} \\
& +2 \cdot \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1g}} - 2 \cdot \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1g}}
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{3g}^r)| + 1 \cdot |(b_{1u}^l) (b_{3g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{3g}^r)| \\
& +1 \cdot |(b_{2g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{2g}^r) (a_u^r)| + 1 \cdot |(b_{2g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{2g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{2u} + b_{3g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{3u} + b_{2g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} - \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} + \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \\
& - \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} - \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \\
& - \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} - \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} + \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} + \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} + \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} - \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} + \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} - \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \\
& + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} - \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} - \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \\
& + \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} + \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}
\end{aligned}$$

summed up:

$$\begin{aligned}
& +2 \cdot \left| \underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}} \right|
\end{aligned}$$

$$e^3 b_{3u} * e^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right|
\end{aligned}$$

dimer ci vectors:

$$\begin{aligned}
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g}
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$+1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{3g}^r) (a_u^r)|$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(+a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| \\
& - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| - \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| \\
& - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

$$e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right|
\end{aligned}$$

dimer ci vectors:

summed up:

$$+2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|$$

or, expressed in terms of symmetry:

monomer determinants:

$$+1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| \\ -1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{3g}^r) (a_u^r)|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| \\ +1 \cdot |(+a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| \\ -1 \cdot |(+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| \\ -1 \cdot |(+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} + \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \\
& + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \\
& + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} + \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} + \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \\
& - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \\
& + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} + \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \\
& - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \\
& + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} + \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} + \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \\
& - \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} - \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} - \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} - \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} + \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} + \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \\
& - \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} - \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}
\end{aligned}$$

summed up:

$$+ \quad 2 \quad \cdot \quad \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \quad - \quad 2 \quad \cdot \quad \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}$$

$$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\
& - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI vectors:

monomer ci vectors:

$$\begin{aligned}
& + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| - \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& - \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\
& + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right|
\end{aligned}$$

dimer ci vectors:

summed up:

$$\begin{aligned}
& -4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right| \\
& + 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| \\
& + 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{b_{1u}} \right| + 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{b_{1u}} \right| \\
& - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{b_{1u}} \right| - 4 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{b_{1u}} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$\begin{aligned}
& +1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{2g}^r)| - 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| \\
& - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{3g}^r) (a_u^r)| - 1 \cdot |(b_{3g}^l) (a_u^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{3g}^r) (a_u^r)|
\end{aligned}$$

substitution by dimer:

$$\begin{aligned}
& +1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| \\
& +1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| - 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| \\
& -1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)|
\end{aligned}$$

multiplied out:

$$\begin{aligned}
& + \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} - \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \\
& + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \\
& - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \\
& + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \\
& - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \\
& - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \\
& + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \\
& - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} + \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} - \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \\
& + \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} - \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} - \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} + \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} - \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \\
& - \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} + \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} + \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} - \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} + \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \\
& \quad \quad \quad - \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} + \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}
\end{aligned}$$

summed up:

$$\begin{aligned}
& -4 \cdot \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} - 4 \cdot \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} + 4 \cdot \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} + 4 \cdot \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} + 4 \cdot \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} + 4 \cdot \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \\
& \quad \quad \quad - 4 \cdot \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} - 4 \cdot \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}
\end{aligned}$$

$$i^3 b_{3u} * i^3 b_{3u}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:

monomer ci vectors:

$$+ \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^1 (b_{2g}^l)^1 (b_{3g}^l)^2 (a_u^l)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^0 (b_{2g}^r)^2 (b_{3g}^r)^1 (a_u^r)^1}_{b_{3u}^r} \right| + \left| \underbrace{(b_{1u}^l)^0 (b_{2g}^l)^2 (b_{3g}^l)^1 (a_u^l)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(b_{1u}^r)^1 (b_{2g}^r)^1 (b_{3g}^r)^2 (a_u^r)^0}_{b_{3u}^r} \right|$$

substitution by dimer:

$$\begin{aligned} & + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ & + \left| \underbrace{(+a_g + b_{1u})^1 (+b_{3u} + b_{2g})^1 (+b_{2u} + b_{3g})^2 (+b_{1g} + a_u)^0}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \\ & + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^1 (-b_{3u} + b_{2g})^1 (-b_{2u} + b_{3g})^2 (-b_{1g} + a_u)^0}_{b_{3u}^r} \right| \\ & + \left| \underbrace{(+a_g + b_{1u})^0 (+b_{3u} + b_{2g})^2 (+b_{2u} + b_{3g})^1 (+b_{1g} + a_u)^1}_{b_{3u}^l} \right| \cdot \left| \underbrace{(-a_g + b_{1u})^0 (-b_{3u} + b_{2g})^2 (-b_{2u} + b_{3g})^1 (-b_{1g} + a_u)^1}_{b_{3u}^r} \right| \end{aligned}$$

dimer ci vectors:

$$\begin{aligned}
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} - \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^1(b_{3g})^1(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^2(a_u)^1}_{a_g} - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^0(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^1(b_{2g})^2(b_{3g})^1(a_u)^1}_{b_{1u}} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^0}_{b_{1u}} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^2(a_u)^1}_{a_g} - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^0}_{a_g} \\
& - \underbrace{(a_g)^1(b_{3u})^2(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^1(b_{3g})^1(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^0}_{a_g} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^1(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^2(a_u)^1}_{b_{1u}} + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^0}_{b_{1u}} \\
& + \underbrace{(a_g)^1(b_{3u})^1(b_{2u})^2(b_{1g})^0(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g} + \underbrace{(a_g)^0(b_{3u})^2(b_{2u})^1(b_{1g})^1(b_{1u})^0(b_{2g})^2(b_{3g})^1(a_u)^1}_{a_g}
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^2 (b_{1g})^0 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^1 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^2 (a_u)^0}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^2 (b_{1g})^1 (b_{1u})^0 (b_{2g})^1 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^1 (b_{3u})^2 (b_{2u})^1 (b_{1g})^0 (b_{1u})^0 (b_{2g})^1 (b_{3g})^2 (a_u)^1}_{a_g} \right| - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^2 (b_{1g})^1 (b_{1u})^1 (b_{2g})^2 (b_{3g})^1 (a_u)^0}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^1 (b_{2u})^1 (b_{1g})^0 (b_{1u})^1 (b_{2g})^2 (b_{3g})^2 (a_u)^1}_{a_g} \right| + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^2 (b_{1g})^0 (b_{1u})^1 (b_{2g})^1 (b_{3g})^1 (a_u)^1}_{a_g} \right| \\
& + 2 \cdot \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^1 (b_{2g})^1 (b_{3g})^2 (a_u)^0}_{a_g} \right| + \left| \underbrace{(a_g)^0 (b_{3u})^2 (b_{2u})^1 (b_{1g})^1 (b_{1u})^0 (b_{2g})^2 (b_{3g})^1 (a_u)^1}_{a_g} \right|
\end{aligned}$$

or, expressed in terms of symmetry:

monomer determinants:

$$+1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{1u}^l) (b_{2g}^l) (b_{3g}^r) (a_u^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{1u}^r) (b_{2g}^r)| + 1 \cdot |(b_{3g}^l) (a_u^l) (b_{3g}^r) (a_u^r)|$$

substitution by dimer:

$$+1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(a_g + b_{1u}) (+b_{3u} + b_{2g}) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)|$$

$$+1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-a_g + b_{1u}) (-b_{3u} + b_{2g})| + 1 \cdot |(b_{2u} + b_{3g}) (+b_{1g} + a_u) (-b_{2u} + b_{3g}) (-b_{1g} + a_u)|$$

multiplied out:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| \\
& - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| + \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| + \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| \\
& + \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| - \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
& + \left| \underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}} \right| + \left| \underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}} \right| - \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& - \left| \underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}} \right| - \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - \left| \underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| \\
& + \left| \underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}} \right| + \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$

summed up:

$$\begin{aligned}
& + \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \cdot \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| - 2 \cdot \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| \\
& - 2 \cdot \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| + 2 \cdot \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| + \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$
